

Wednesday Warm Up: Resistors in Series/Parallel, Kirchhoff's Rules

Prof. Jordan C. Hanson

February 26, 2025

1 Memory Bank

- $\Delta V = IR$... Ohm's Law.
- $R_{\text{tot}} = R_1 + R_2 + \dots$... Resistors *in series*
- $R_{\text{tot}}^{-1} = R_1^{-1} + R_2^{-1} + \dots$... Resistors *in parallel*
- Kirchhoff's Rule #1: Total current into a junction equals total current out of a junction.
- Kirchhoff's Rule #2: The total change in voltage around a loop is zero.

2 Resistors in Series/Parallel, Kirchhoff's Rules

1. Consider Fig. 1. (a) If $V_0 = 12 \text{ V}$, and the resistance of bulb B is $R_B = 5 \Omega$, what current flows if the switch in the middle is open? (b) If $R_A = R_B$, what is the total resistance if the switch is closed? (c) What current flows if the switch is closed?

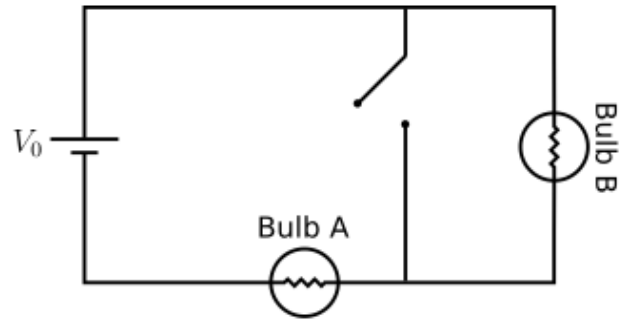


Figure 1: Two light bulbs connected to an EMF V_0 .

3. Repeat the previous two exercises, but let $V_0 = 5 \text{ V}$, $R_A = 4 \Omega$, and $R_B = 8 \Omega$.
4. (a) What would the power consumption be if a third bulb was added in parallel with the first two, and each had a resistance of 5Ω ? (b) Argue that the result makes sense in light of the conservation of energy. (c) What current would flow if the three bulbs were instead connected in series? (d) What would be the power consumption in that case?
2. What is the power consumption if the switch is (a) open, (b) closed?